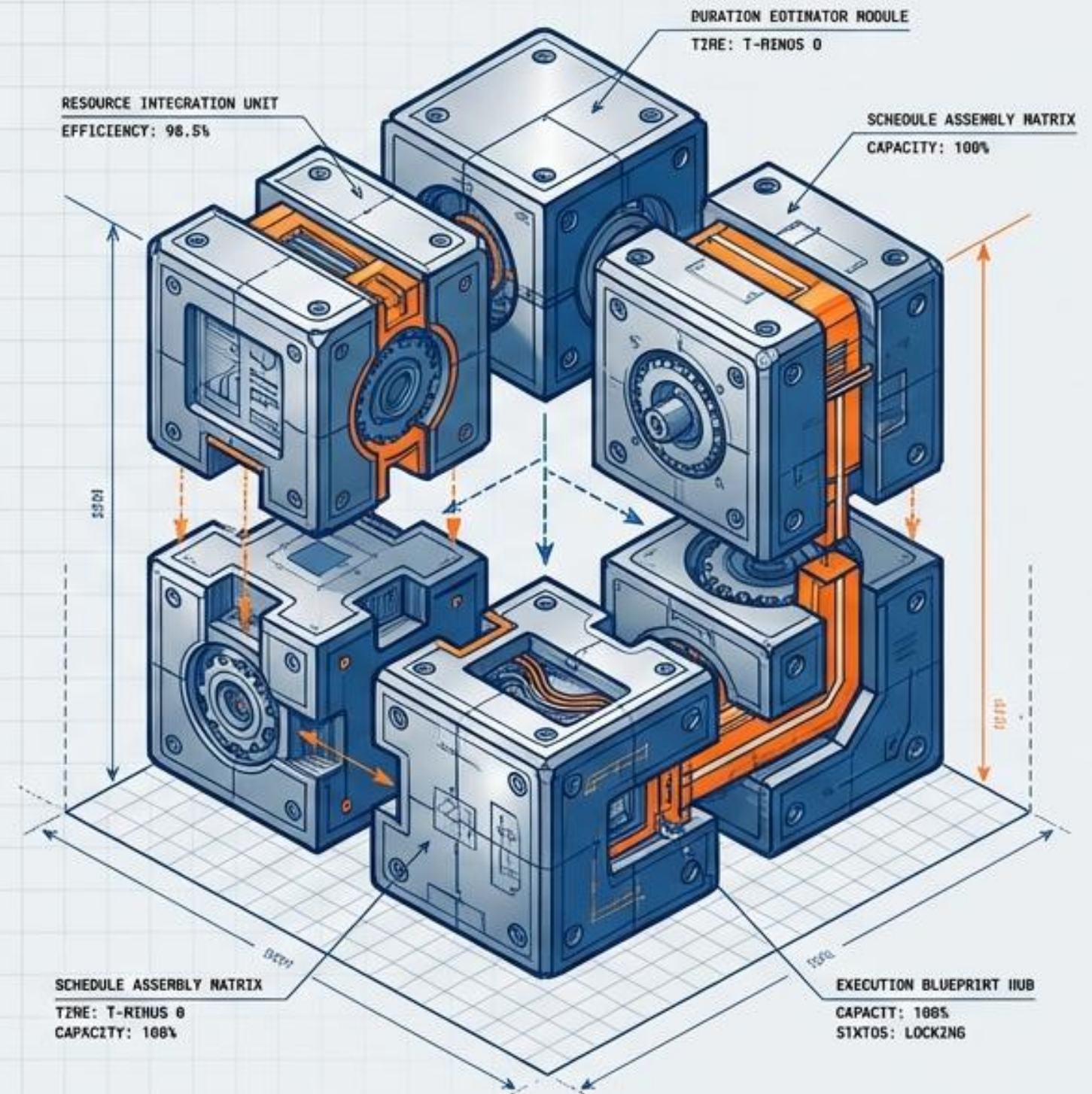


Certyfikat PMP: Project resources management

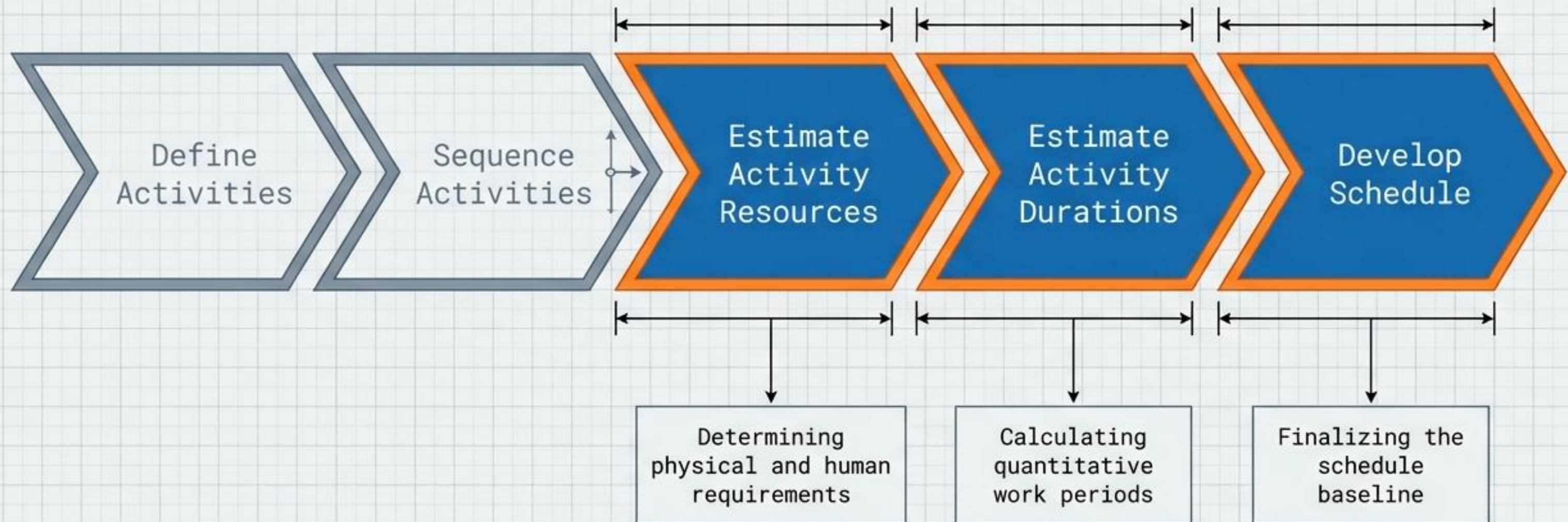
dr Zbigniew Galar

The Blueprint of Execution:
Estimating Resources, Durations,
and the Project Schedule



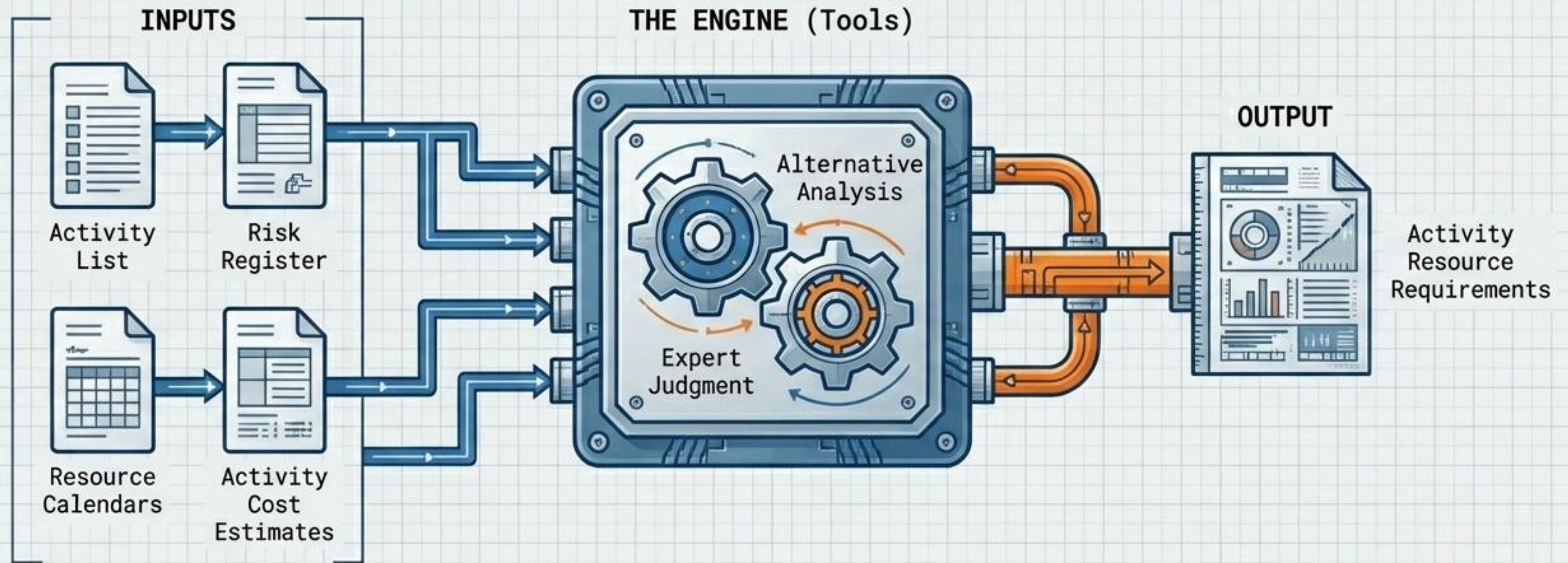
The Planning Engine

The critical sequence of PMP planning processes.



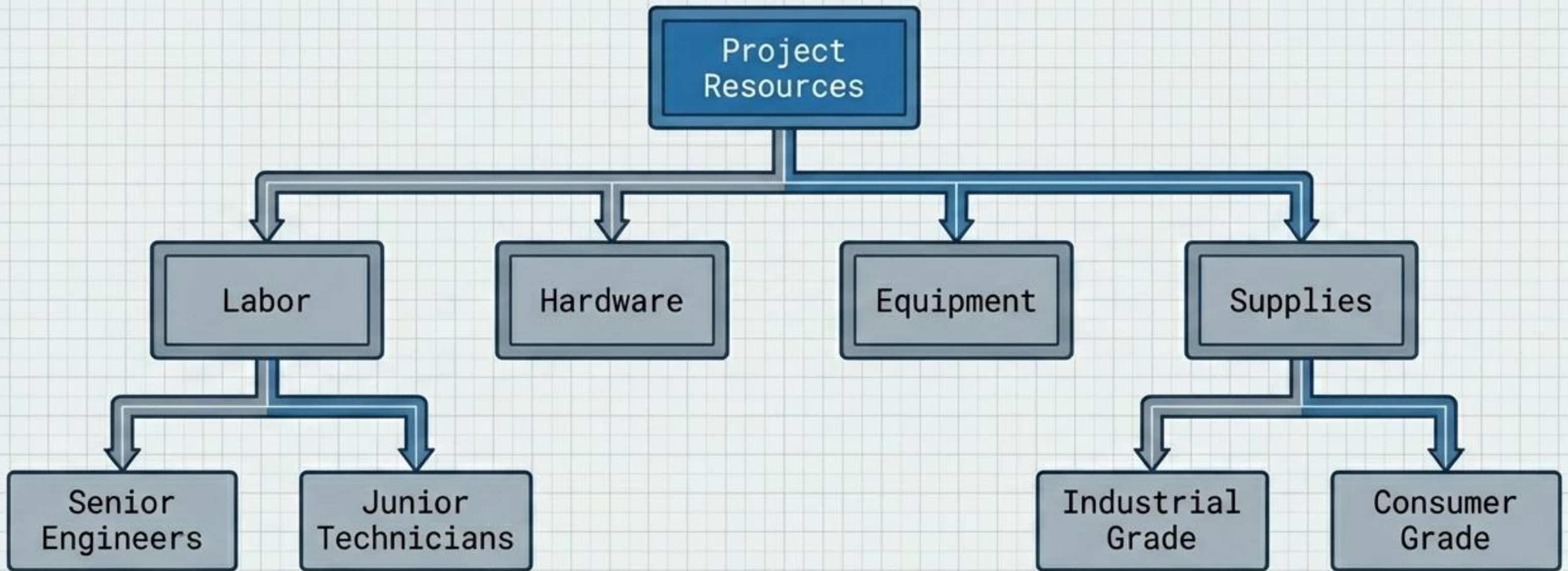
Process 1: The Resource Estimation Engine

Resources extend beyond human capital—they include equipment, materials, supplies, hardware, and software. Raw project data must be processed to determine precise requirements.



The Resource Breakdown Structure (RBS)

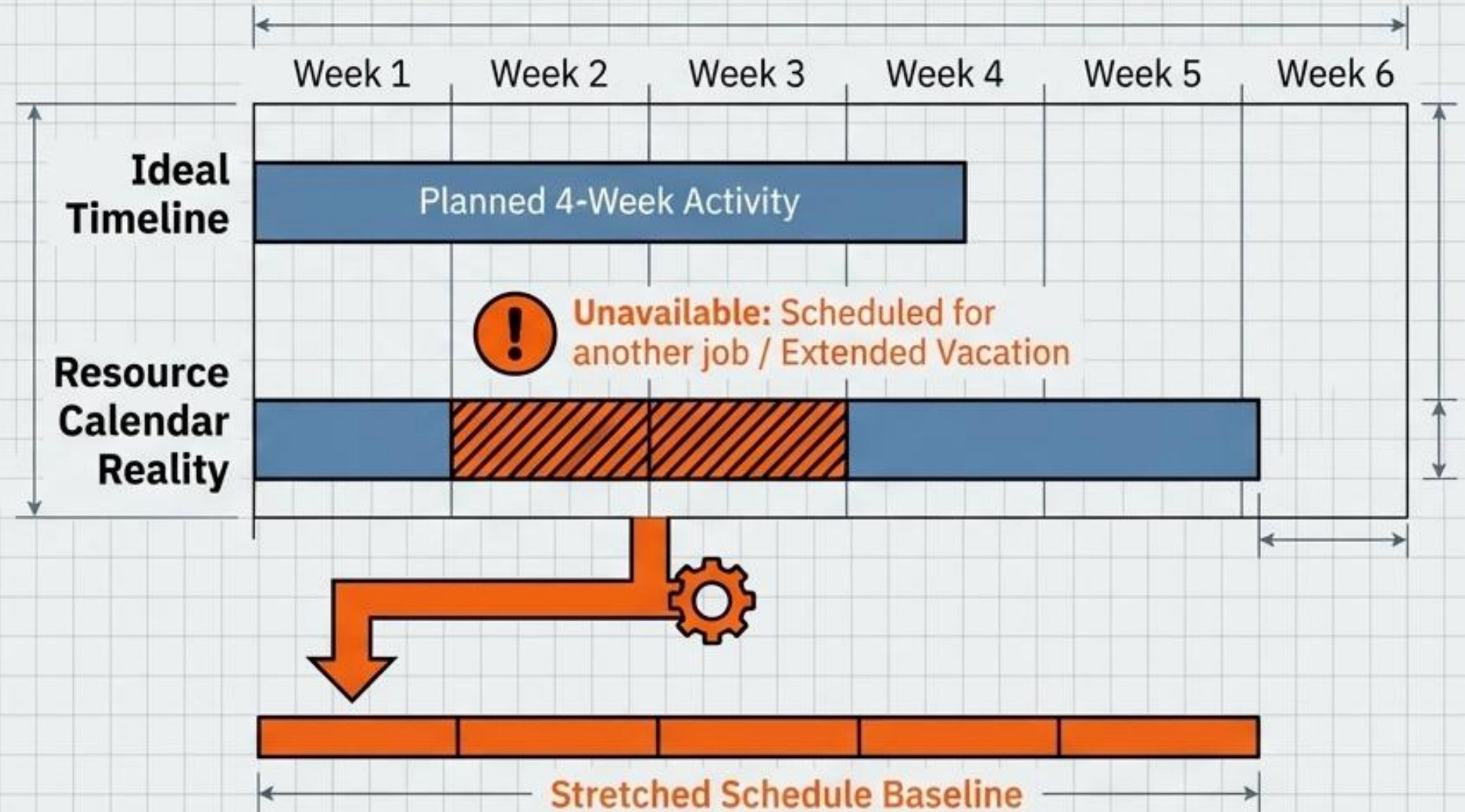
The RBS lists resources by category and type, allowing the project manager to precisely identify quantities needed for each individual work package.



The Reality Check: Resource Calendars

Resource availability dictates project reality.

Resource calendars track the specific time frames, quantities, and capabilities of human and material resources. They act as a hard constraint on schedule development.



Process 2: Estimate Activity Durations

Duration cannot be estimated until Resources are assigned. The quantity and skill level of the resource strictly dictates the speed of the work.

The House-Painting Example

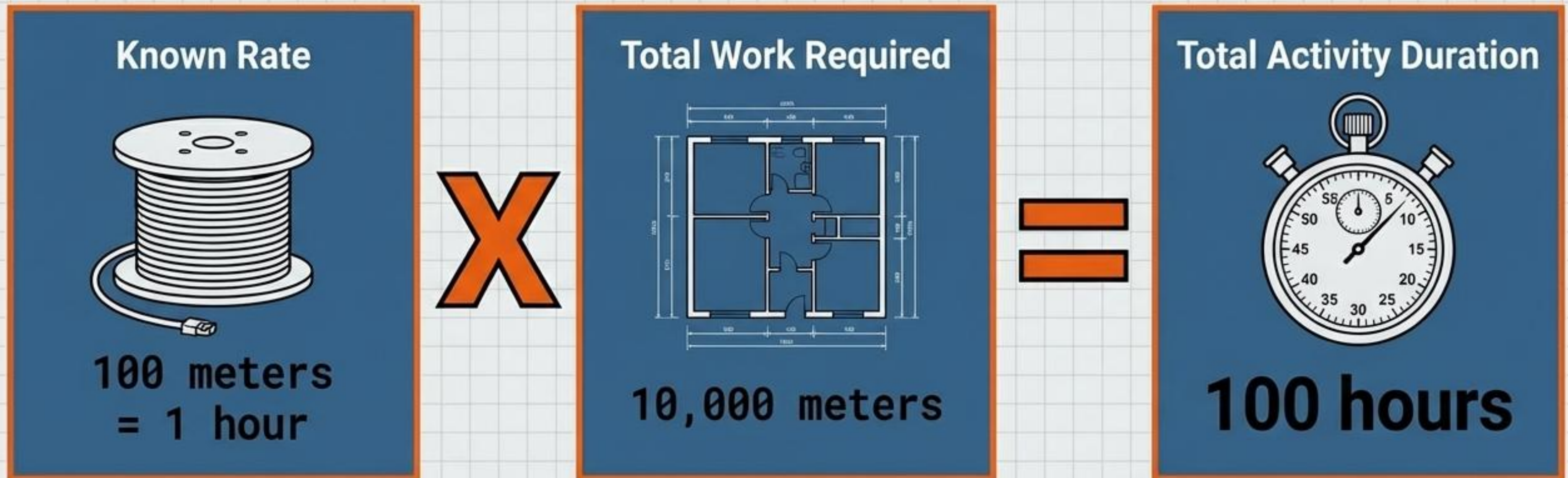
- **Task requirement:** 3 days to prime and dry.
- **Schedule constraint:** Work begins Saturday, but the crew does not work on Sunday.
- **Calculation:** 3 days work + 1 non-work day = 4 days total Activity Duration.



Matrix: The Estimating Arsenal

Analogous (Top-Down)	Parametric	Bottom-Up
Form of expert judgment using historical data	Algorithm using statistical relationships	Roll-up of individually estimated granular tasks
Low Cost / Low Time	Medium Cost / Medium Time	High Cost / High Time
Less Accurate (Gross Value). Best for early project phases.	Highly Accurate if historical baseline data is reliable.	Most Accurate. Used when lower-level detail is fully known.

Parametric estimation thrives on quantifiable historical metrics.



$$[\text{Known Rate}] \times [\text{Total Quantity}] = [\text{Total Duration}]$$

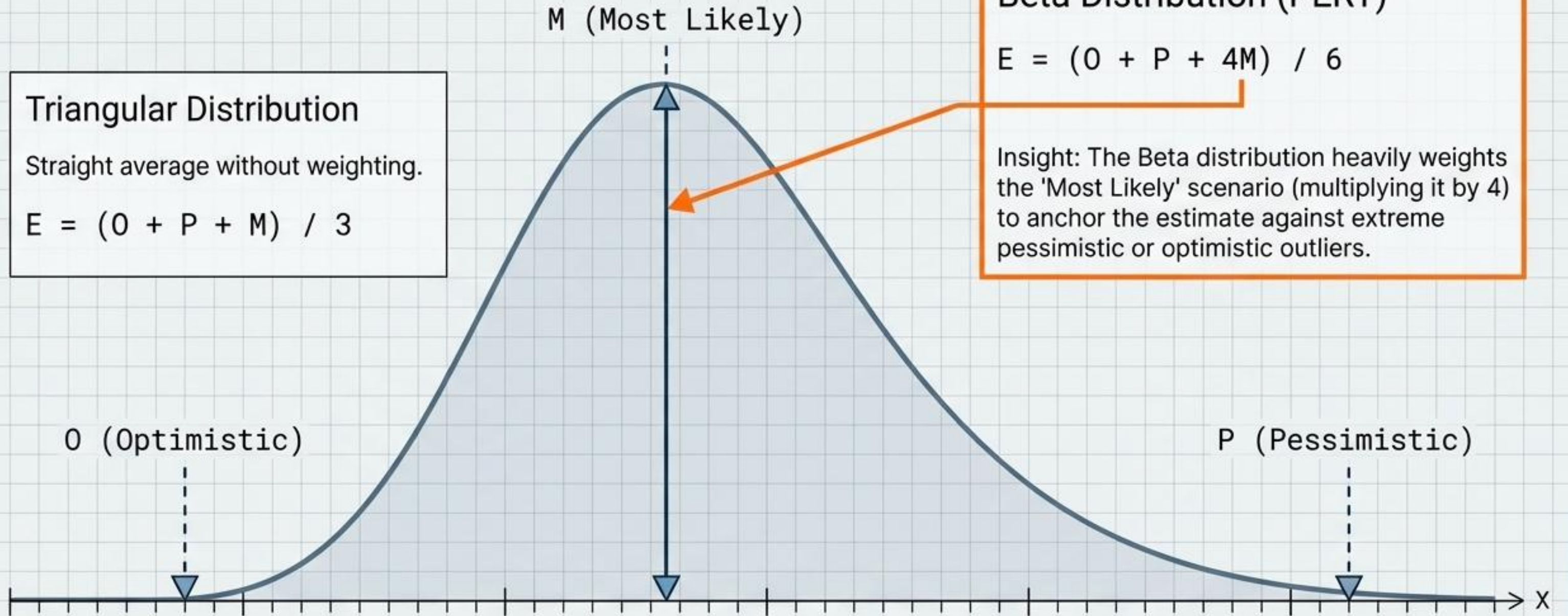
Three-Point Estimating Formulas (PERT)

Triangular Distribution
Straight average without weighting.
$$E = (O + P + M) / 3$$

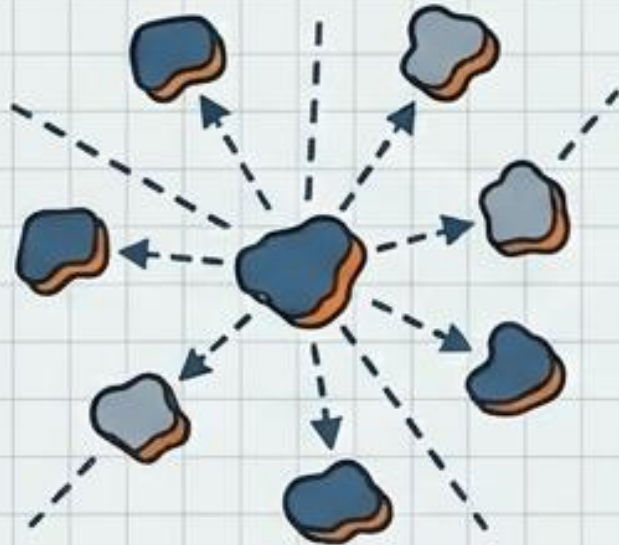
Beta Distribution (PERT)

$$E = (O + P + 4M) / 6$$

Insight: The Beta distribution heavily weights the 'Most Likely' scenario (multiplying it by 4) to anchor the estimate against extreme pessimistic or optimistic outliers.



Human Factors in Estimation



Brainstorming

- Open forum environment.
- No idea is a bad idea.
- Designed to prevent judging contests.



Delphi Technique

- Anonymous subject matter experts provide estimates to prevent bias.
- Feedback is circulated continuously.
- Process repeats until exact consensus is reached.

Real World Scenario: Desert State University (DSU)

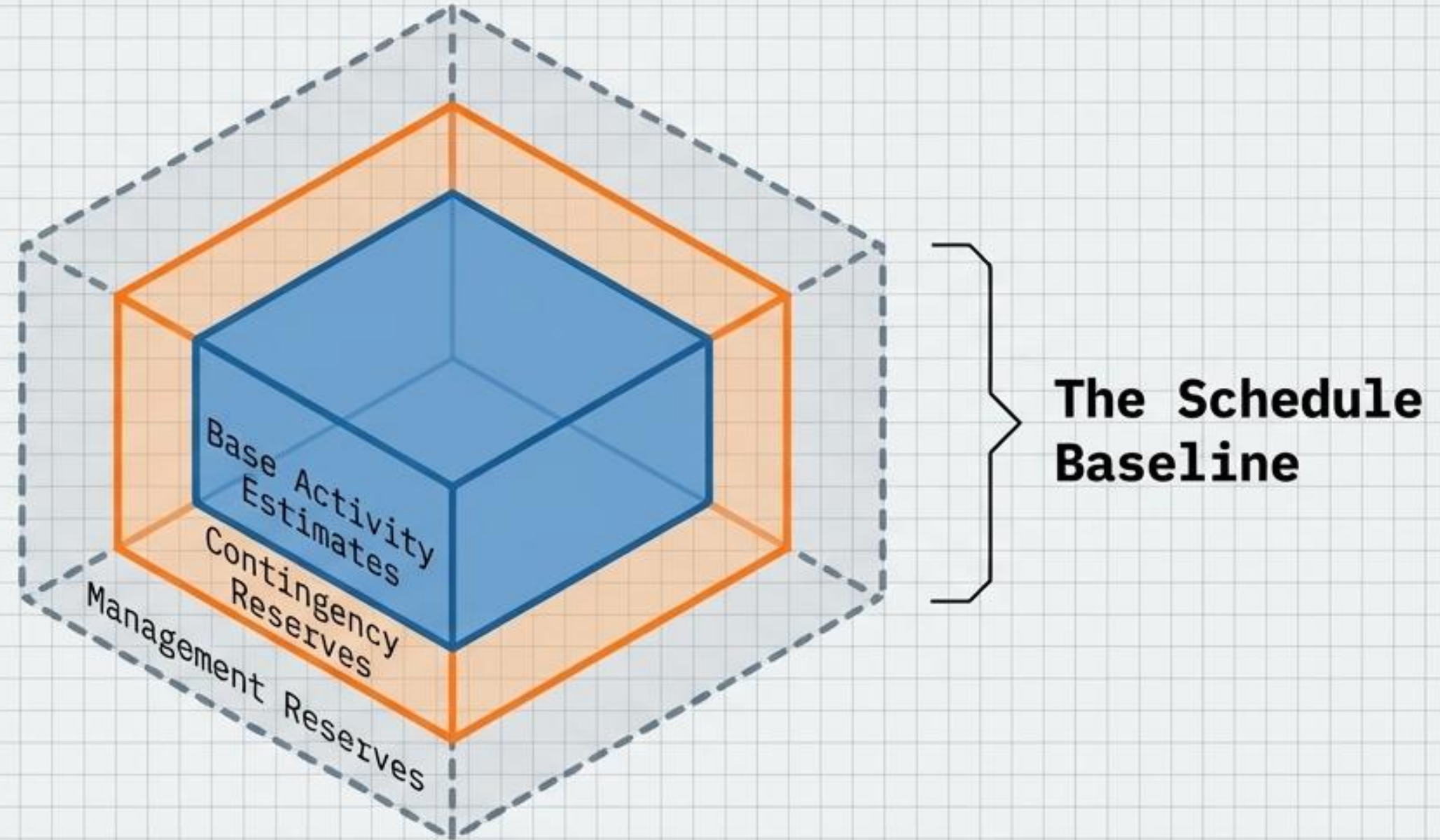
- Mike Walter (Mainframe): Relies on sole Expert Judgment from past interface programs.
- Kate Langdon (Testing): Blends Historical Information (project binders) with Expert Judgment (lead tester consultation).

Matrix: Reserve Analysis

How project managers pad schedules scientifically.

Contingency Reserves	Management Reserves
Risk Type: Known-Unknowns (documented risks with mitigation plans).	Risk Type: Unknown-Unknowns (unforeseen events within project scope).
Baseline Status: Included in the schedule baseline.	Baseline Status: Not included in the schedule baseline (requires a baseline change to use).
Example: Adding 10 hours to a 100-hour estimate for potential structural hurdles. Final activity duration = 110 hours.	Example: Time set aside for the 'funny feeling' that something entirely unseen will occur.

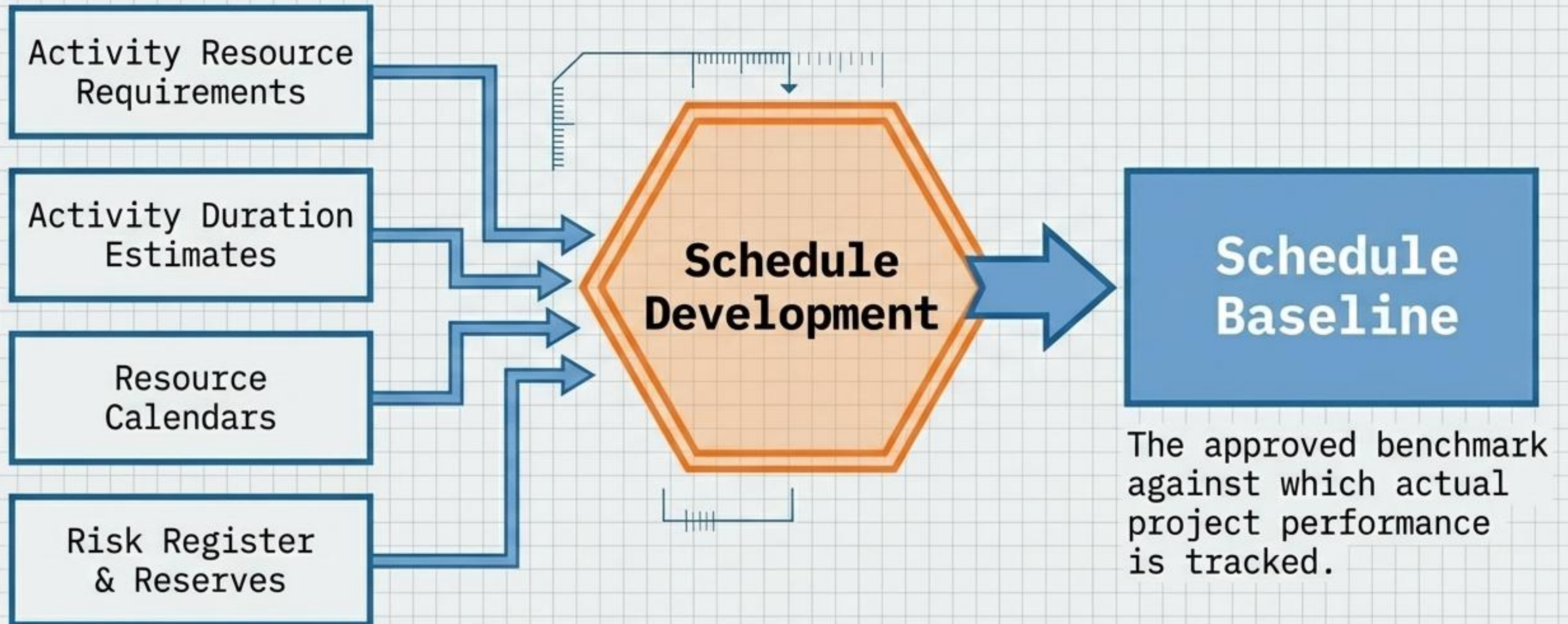
Visualizing the Schedule Baseline



Key Insight: Management reserves are held strictly outside the approved baseline but remain within the overall project scope.

Process 3: Develop Schedule

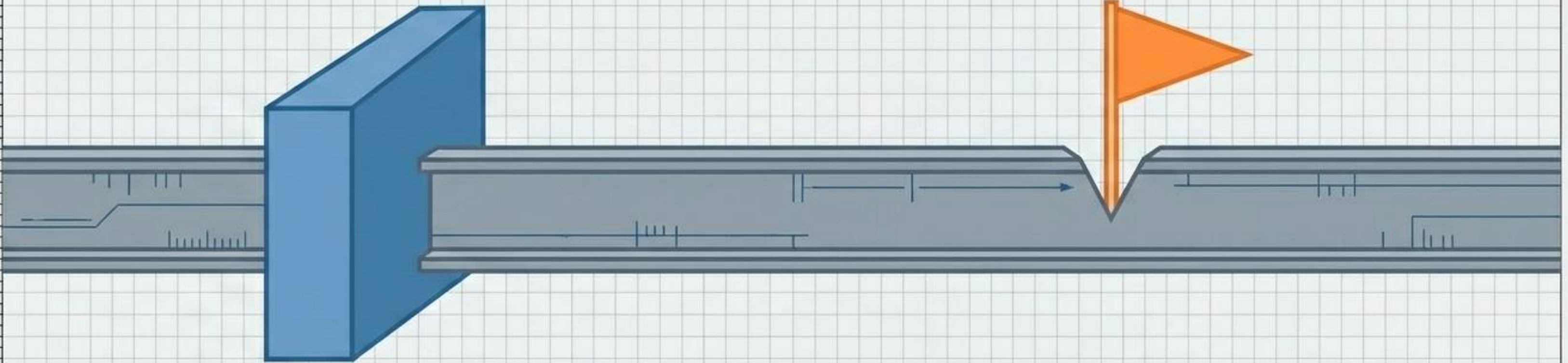
The heart of the planning process. An iterative cycle finalizing start/finish dates and activity sequences.



Navigating Schedule Constraints

Imposed Date

Key Event / Milestone



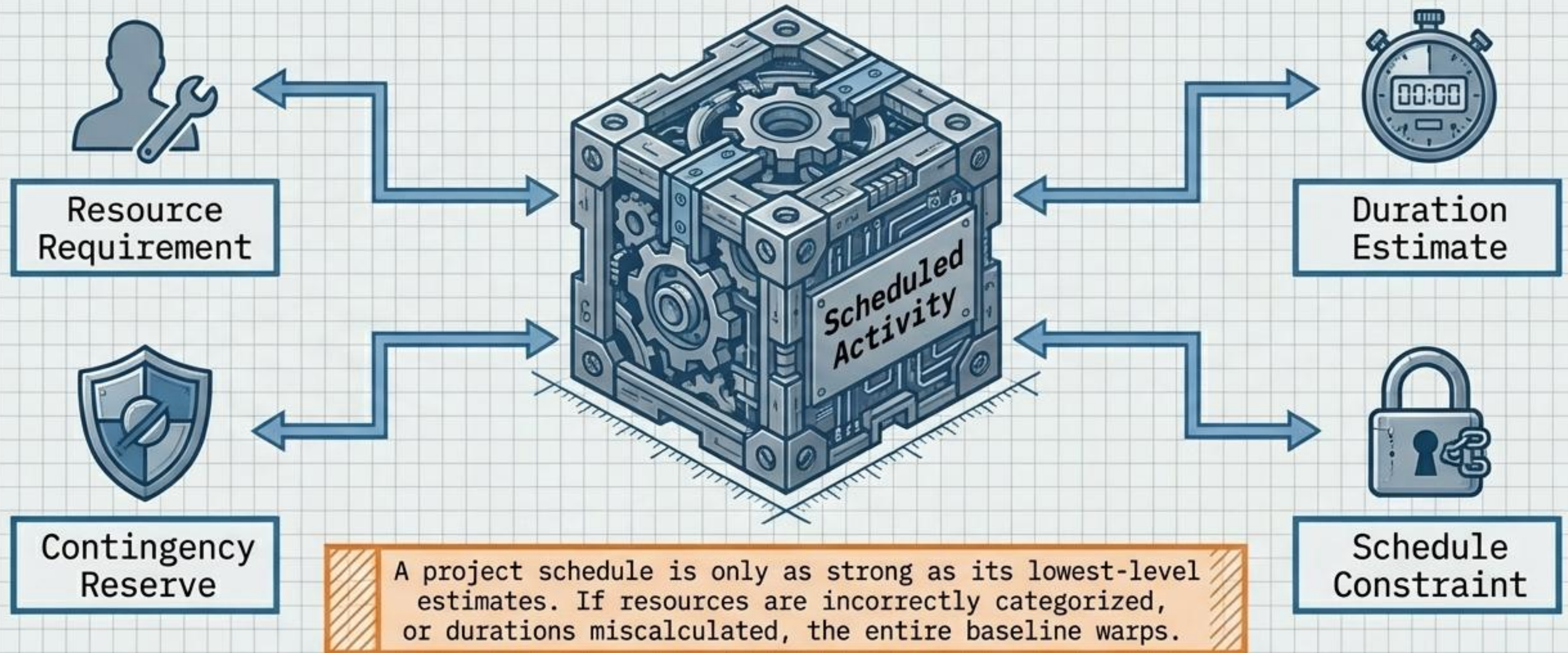
Systemic restrictions.

Example: Painting cannot start until primer dries. If primer finishes Wednesday, painting has an imposed "start no earlier than" constraint of Thursday.

Specific deliverables requested by stakeholders by a specific date.

Once verbally agreed upon, they often become inflexible, cast-in-stone constraints.

The Anatomy of a Project Plan



Accurate planning is the ultimate investment for project success. The Schedule Baseline reflects exactly what we know at this specific moment in time.